

# Deep Learning

Vazgen Mikayelyan

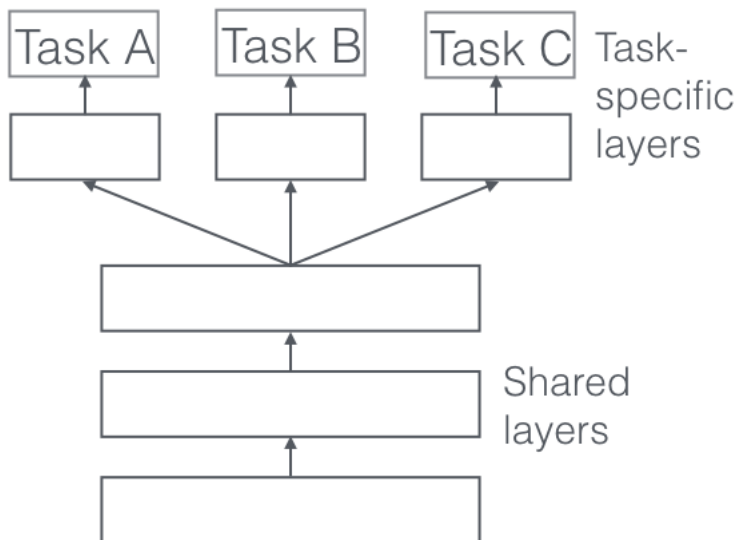


## 1 Multi-Task Learning

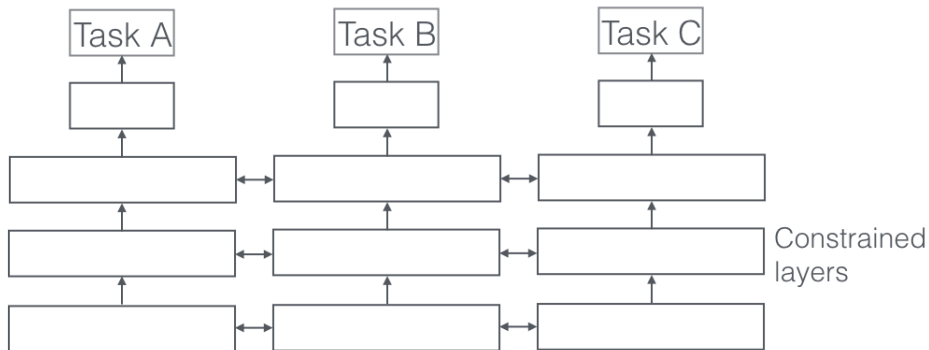
## 2 Metrics for Classification Problems

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- Training on a set of tasks that could benefit from having shared lower level features.
- Amount of data you have for each task is quite similar.
- Can train a big enough neural network to do well on all the tasks.

# Outline

1 Multi-Task Learning

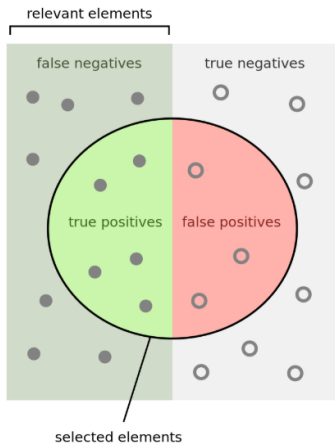
2 Metrics for Classification Problems

## Definition 1

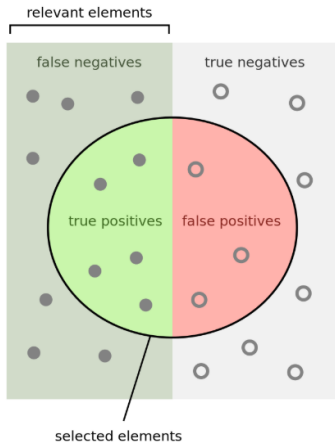
*Accuracy in classification problems is the percent of correct predictions made by the model over all kinds predictions made:*

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total numbers of predictions made}}$$

# Precision and Recall



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How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

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**Answer:** If  $Precision = 0.1$  and  $Recall = 0.95$ , then their mean is equal to 0.525 and  $F1 \approx 0.18$ .

# ROC Curve

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- The diagnostic performance of a test or the accuracy of a test to discriminate diseased cases from normal cases is evaluated using ROC curve analysis.
- A ROC curve is a way to compare diagnostic tests. It is a plot of true positive rate against the false positive rate:

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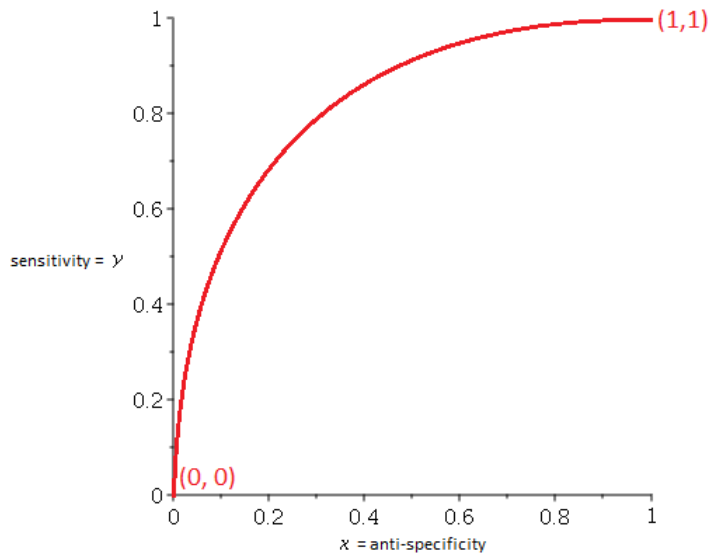
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- What if  $AUC \approx 0.5$ ?
- What if  $AUC \approx 1$ ?
- How to choose threshold?
- How to use ROC curve for multi-class model?